SCAMSHIELD

ScamShield Hackathon

A focused, demo-ready prototype for a six-hour build

SIX-HOUR MVP

Product concept and implementation plan

Prepared as a two-track project: a six-hour hackathon demonstrator and a scalable long-term fraud-protection platform.

Design principle: build a focused first slice without discarding the larger product vision.

1. Executive summary

ScamShield Hackathon is a small, demonstrable version of a larger fraud-protection platform. It helps a user inspect suspicious messages, see the warning signs, submit community reports and view an illustrative scam-intelligence feed.

The goal is not to claim that a six-hour prototype can detect every scam. The goal is to demonstrate a useful, explainable workflow and show how the same core can grow into mobile protection, audio analysis and approved social-platform integrations.

- Input: a pasted message, website address or payment request.
- Analysis: transparent starter rules identify common warning signs.
- Output: risk band, detected signals and recommended next steps.
- Community reports: sample and locally submitted reports appear in the demo feed.
- Social bot preview: shows how a future bot could warn about suspicious group messages.

2. Features included in the prototype

- **Message and link analyser:** accepts text or a URL and highlights patterns such as requests for OTPs, urgency, prize claims, upfront fees and links.
- **Explainable results:** shows the reasons behind the assessment instead of displaying only a score.
- **Community reporting:** lets the presenter add a report to browser-local storage.
- **Intelligence feed:** displays sample reports and reports submitted during the demo.
- **Social bot preview:** demonstrates a possible warning response without connecting to a real social platform.
- **Responsive dashboard:** provides a polished interface for a laptop or mobile browser.

3. Six-hour build schedule

- **Hour 1 — Setup and demo story:** run the prototype, assign roles, agree on the target judging problem and prepare two example messages.
- **Hour 2 — Detection quality:** add realistic test cases, improve the rules and verify that legitimate messages do not trigger too many warnings.
- **Hour 3 — Backend (optional):** if the team is comfortable, add a small API and persistent database. Do not risk the whole demo for a complex deployment.
- **Hour 4 — AI integration (optional):** connect a language-model API if credentials, network access and time are available. Keep rule-based fallback behaviour.
- **Hour 5 — Testing and polish:** check errors, mobile layout, empty inputs and explanation quality.
- **Hour 6 — Rehearsal:** reset demo data, prepare a short pitch and test the full presentation from beginning to end.

4. Suggested demo script

- Open the app and load a message claiming an account will be suspended unless the recipient shares an OTP.
- Run the analysis and explain the detected urgency and credential request.
- Explain the recommended action: do not share the code; verify through an official channel.
- Submit a community report and show it in the feed.
- Show the social bot preview and explain that a real connector would require approved platform access.

- Finish by showing how multiple reports could eventually be connected into a fraud-campaign graph.

5. What is not live yet

- The starter analyser uses simple pattern rules; it is not a trained AI model and its score is not a calibrated probability.
- The sample threat feed is not connected to live threat-intelligence providers.
- Reports are stored only in the current browser and are not shared across devices.
- The social bot is a simulation; no social platform is connected.
- There is no automatic SMS reading, live call recording, payment blocking or guaranteed scam attribution.

6. Evaluation checklist

Test	What to check
Suspicious message	Flags OTP/password requests, urgency or upfront-fee patterns with clear explanations.
Legitimate message	Avoids overstating certainty; shows that no detected pattern does not prove safety.
Empty input	Displays a helpful prompt rather than an error.
Report submission	Adds the report to the feed and updates counts.
Demo recovery	Refresh the page and confirm sample data still loads; explain that local data is browser-specific.

Safety note: do not enter real passwords, PINs, OTPs or sensitive private conversations into the prototype. Treat user reports as unverified until corroborated.